

Genetic report for longevity

Allergies, Asthma and Atopy

Asthma, allergic rhinitis, IgE and eosinophils, nasal polyps, food and drug allergy, histamine

Sérgio F · Report date: 02-10-2026

Atopy is the tendency for immune responses of type 2 (IgE, eosinophils, IL-4/IL-5/IL-13) to harmless allergens. With age, asthma and rhinitis tend to change, and inflammation of the airways adds up to inflammation. This report reads the genes of asthma (17q21 ORMDL3/GSDMB, IL33, IL1RL1, TSLP, HLA-DQ), allergic rhinitis, IgE and eosinophils, nasal polyps, food allergy and medication, response to salbutamol (ADRB2) and histamine degradation (AOC1, HNMT). The eczema is already in the report Nails, Cichtarization and Inflammation of the Skin.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

Contents

• Summary	3
• Asthma	4
• Allergic rhinitis, nasal polyps and sinusitis	6
• IgE, eosinophils and food allergy	7
• Histamine and drug response.....	9
• Mechanisms of ageing.....	10
• Practical part	11
• Limitations.....	12
• Methods and sources	13

Summary

23

variants

18

in the chip

3

imputed
($DR2 \geq 0.8$)

2

not found

13

Scores with
percentile

Overall reading. The genetics of atopy is favourable, with one exception in sinusitis. In asthma: the large score (22 679 variants) is at the 4th percentile, that of 212 loci in 31 and that of early childhood asthma in 66; the onset asthma in adults is in 45. 17q21 (ORMDL3/GSDMB) is heterozygous, TSLP rs1837253 has 2 risk alleles (CC) and SMAD3 none (AA). In the atopic disease in general: the 14th percentile in the atopy score, 22 in the allergic disease score and 10 in the eczema score (matches with the Nails report, Wound healing and Skin Inflammation); allergic rhinitis in 52; FLG R501X was not read with quality here, but the §12 profile and the Skin Care report did not find common FLG mutations. In IgE and eosinophils: FCER1A TT and EMSY GG without risk alleles Genetic IgE in 46 and eosinophils in 58; food allergy in 33. The weak point is chronic sinusitis: the score of chronic sinusitis with polyps is at the 96th percentile, although the PGS of nasal polyps is in the 39; it is worth to be aware of persistent nasal congestion or loss of smell. In histamine, 1 allele in each of the two variants of DAO (AOC1) and 1 in HNMT: slightly slower degradation without meaning without symptoms (details in the §12 profile, including tryptase and mast cells).

Favourable

- **Asthma (22 679 variants):** the 4th percentile
- **Atopic disease in general (124 loci):** the 14th percentile
- **Allergic disease: asthma, rhinitis or eczema (136 loci):** the 22nd percentile
- **Atopic eczema (71 loci):** the 10th percentile
- **SMAD3 rs744910 AA:** SMAD3: TGF- β pathway, airway remodeling.
- **C11orf30/EMSY rs2155219 GG:** 11q13.5 (EMSY/LRRC32): shared locus of rhinitis, asthma and eczema.
- **FCER1A rs2251746 TT:** FCER1A: high affinity IgE receptor; the main locus of total IgE.

Unfavourable

- **Chronic sinusitis:** the 96th percentile
- **TSLP rs1837253 CC:** TSLP: epithelial alarmin (target of tezepelumab); T allele protects from asthma and rhinitis.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 18 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Asthma

The locus 17q21 (ORMDL3/GSDMB) is the strongest of early childhood asthma; IL33, IL1RL1 (its ST2 receptor), TSLP and HLA-DQ bind the epithelium to the 2 type immunity and are also valid for adult asthma.

Ageing mechanisms: Chronic Inflammation; Intercellular communication alteration; Remodelling of the extracellular matrix

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Asthma (22 679 variants)	Favourable	the 4th percentile · z=-1.81	Higher score : higher risk. PGS001849 (PGS Catalog; Penalized regression (bigstatsr); Asthma; PGP000263, PMID 34995502). Variants with genotype: 21984 of 22679 (covering of 96.7% coefficients).
Asthma (212 loci)	Typical / intermediate	the 31st percentile · z=-0.51	Higher score : higher risk. PGS004252 (PGS Catalog; Genome-wide signifier SNPs; Asthma; PGP000544, PMID 37080133). Variants with genotype: 197 of 212 (covering of 89.1% coefficients).
Early childhood asthma (105 loci)	Typical / intermediate	the 66th percentile · z=0.37	Higher score : higher risk. PGS002248 (PGS Catalog; PRSice (C+T); Childhood asthma; PGP000275, PMID 35055391). Variants with genotype: 99 of 105 (covering of 94.1% coefficients).
Asthma of onset in adults	Typical / intermediate	the 45th percentile · z=-0.10	Higher score : higher risk. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 22 of 24 loci with genotype. Studies: Dahlin A (Adult asthma); Daya M (Adult asthma); Ferreira MAR (Asthma) Pividori M (Asthma (adult onset)). Variants with genotype: 22 of 24 (covering of 89.3% coefficients).

Favourable

- Asthma (22 679 variants) (the 4th percentile)
- SMAD3 rs744910 AA

Unfavourable

- TSLP rs1837253 CC

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SMAD3	rs744910 15:67154447 · G/A/T	AA	chip	G (0; 48%)	moderate	SMAD3: TGF-β pathway, airway remodeling. Meaning: GWAS Catalog: Asthma (Moffatt MF 2010; p=4e-09); G ↑ allele

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TSLP	rs1837253 5:111066174 · T/C	CC	chip	C (2; 75%)	strong	TSLP: epithelial alarmin (target of tezepelumab); T allele protects from asthma and rhinitis. Meaning: GWAS Catalog: Asthma (Zhou W 2022; p=4e-104); allele C ↑

Typical / intermediate variants (6)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ORMDL3/GSDMB	rs7216389 17:39913696 · C/T	CT	chip	T (1; 48%)	strong	17q21: T allele increases the expression of ORMDL3/GSDMB and the risk of early childhood asthma (OR ~1.4).
IL1RL1	rs1420101 2:102341256 · C/T	CT	chip	T (1; 34%)	strong	IL1RL1 (ST2, IL-33 receptor): locus of asthma and eosinophils. Meaning: GWAS Catalog: eosinophil (absolute count, mean,

						inv-norm transformed) (Verma A 2024; p=8e-257); allele C ↓
HLA-DQ	rs9273349 6:32658092 · T/C/G	TC	chip	C (1; —)	moderate	HLA-DQB1/DQA1: presentation of allergens; asthma locus. Meaning: GWAS Catalog: Asthma (Moffatt MF 2010; p=7e-14); C ↑ allele
IL13	rs20541 5:132660272 · A/C/G/T	GG	chip	A (0; 21%)	moderate	R130Q: Allele A (130Q) increases IL-13 activity and the risk of asthma and eczema.
IL2RA	rs61839660 10:6052734 · C/T	CC	chip	T (0; 7%)	moderate	IL2RA: regulatory T cells. Meaning: GWAS Catalog: Allergic disease (asthma, hay fever or eczema) (Johansson A 2019; p=3e-28); T ↑ allele
RORA	rs11071559 15:60777789 · C/T	CC	chip	C (2; 87%)	moderate	RORA: 2 type innate lymphoid cells. Meaning: GWAS Catalog: Asthma (Han Y 2020; p=2e-27); allele C ↑

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IL33	rs3939286 9:6210099 · T/ A/C	CT	imputation DR2=1.00	—	strong	IL33: epithelial alarmin; strong locus of asthma and eosinophils.
GSTP1	rs1695 11:67585218 · A/G/T	AG	chip	—	weak	GSTP1 Ile105Val: interaction with pollution (ozone, particles) in asthma in small studies. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ZBTB10	— — · —	—	SNP not found in the data	—	moderate	ZBTB10: locus of asthma and rhinitis. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Allergic rhinitis, nasal polyps and sinusitis

Allergic rhinitis shares most loci with asthma and eczema. Nasal polyps and chronic sinusitis are mucosal type 2 inflammation; respiratory disease exacerbated by aspirin together with asthma, polyps and intolerance to NSAID.

Ageing mechanisms: Chronic Inflammation; Dysbiosis

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Allergic rhinitis (349 variants)	Typical / intermediate	the 52nd percentile · z=0.08	Higher score : higher risk. PGS001259 (PGS Catalog; snpnet; Hayfever/allergic rhinitis; PGP000244, PMID 35324888). Variants with genotype: 302 of 348 (covering of 85.1% coefficients).
Nasal polyps (1783 variants)	Typical / intermediate	the 39th percentile · z=-0.29	Higher score : higher risk. PGS001848 (PGS Catalog; Penalized regression (bigstatsr); Nasal polyps; PGP000263, PMID 34995502). Variants with genotype: 1724 of 1783 (covering of 97.0% coefficients).
Chronic sinusitis	Unfavourable	the 96th percentile · z=1.71	Higher score : higher risk. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 33 of 33 loci with genotype. Studies: Saarentaus EC (Chronic rhinosinusitis with nasal polyposis). Variants with genotype: 33 of 33 (covering of 100.0% coefficients).

Favourable

- C11orf30/EMSY rs2155219 GG

Unfavourable

- Chronic sinusitis (the 96th percentile)

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
C11orf30/EMSY	rs2155219 11:76588150 · G/T	GG	chip	T (0; 49%)	moderate	11q13.5 (EMSY/LRRC32): shared locus of rhinitis, asthma and eczema. Meaning: GWAS Catalog: Self-reported allergy (Hinds DA 2013; p=2e-19); allele T ↑

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
LTC4S	rs730012 5:179793637 · A/C/G/T	AC	chip	C (1; 31%)	weak	LTC4S -444A>C: C allele increases the synthesis of cisteinyl leukotrienes; associated with aspirin-induced asthma in small studies.
ALOX15	rs34210653 17:4632019 · G/A	GG	chip	G (2; 99%)	moderate	ALOX15: lipoxigenase; the T560M variant protects from nasal polyps. Meaning: GWAS Catalog: Nasal polyps (Kristjansson RP 2019; p=8e-27); G ↑ allele

IgE, eosinophils and food allergy

FCER1A (high affinity receptor of IgE), IL13 and HLA region of class II explain part of the total IgE; eosinophil counts are very hereditary. Food allergy has few loci (HLA-DQ, FLG, SERPINB).

Ageing mechanisms: Immunosenescence; Intercellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Atopic disease in general (124 loci)	Favourable	the 14th percentile · z=-1.12	Higher score : higher risk of asthma, rhinitis or eczema. PGS003458 (PGS Catalog; Genome-wide signifier SNPs; General atopic disease; PGP000451, PMID 36804406). Variants with genotype: 116 of 124 (covering of 89.9% coefficients).
Allergic disease: asthma, rhinitis or eczema (136 loci)	Favourable	the 22nd percentile · z=-0.74	Higher score : higher risk. PGS000810 (PGS Catalog; Variantly associated with allergic disease (asthma, hay fever or eczema); Allergic disease (asthma, hay fever or eczema); PGP000195, PMID 29083406). Variants with genotype: 126 of 136 (covering of 91.0% coefficients).
Atopic eczema (71 loci)	Favourable	the 10th percentile · z=-1.26	Higher score : higher risk. PGS003486 (PGS Catalog; Genome-wide signifier SNPs; Atopic eczema; PGP000451, PMID 36804406). Variants with genotype: 70 of 71 (covering of 97.1% coefficients).
Eosinophils (607 loci)	Typical / intermediate	the 58th percentile · z=0.20	Higher score = more eosinophils. PGS000165 (PGS Catalog; Genome-wide signifiers; Eosinophil count; PGP000078, PMID 32888494). Variants with genotype: 486 of 607 (covering of 82.6% coefficients).
Total IgE (7 loci)	Typical / intermediate	the 46th percentile · z=-0.08	Higher score = IgE. Very few loci. PGS000315 (PGS Catalog; GWAS-meaning (lead and secondary) variants; Immunoglobulin E (IgE); PGP000092, PMID 32527150). Variants with genotype: 7 of 7 (covering of 100.0% coefficients).
Food allergy (peanut and others)	Typical / intermediate	the 33rd percentile · z=-0.50	Higher score : higher risk. Few loci. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 25 of 28 loci with genotype. Studies: Hong X (Peanut allergy); Maier L (Doctor-diagnosed food allergy) Maier L (Doctor-diagnosed food allergy plus specific sensitivity) Maier L (Doctor-diagnosed food allergy plus specific sensitivity); Maier L (Doctor-diagnosed food allergy); Maier L (Self-reported food allergy) Maier L (Self-reported food allergy); Marenholz I (Food allergy). Variants with genotype: 25 of 28 (covering of 91.0% coefficients).

Favourable

- Atopic disease in general (124 loci) (the 14th percentile)
- Allergic disease: asthma, rhinitis or eczema (136 loci) (the 22nd percentile)
- Atopic eczema (71 loci) (the 10th percentile)
- FCER1A rs2251746 TT

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FCER1A	rs2251746 1:159302270 · T/C	TT	chip	C (0; 30%)	moderate	FCER1A: high affinity IgE receptor; the main locus of total IgE. Meaning: GWAS Catalog: IgE levels (Granada M 2011; p=5e-26); C ↑ allele

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
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IL5	rs2706334 5:132554779 · C/T	TT	imputation DR2=1.00	C (0; 20%)	moderate	IL5/RAD50 (5q31): region of cytokines type 2; eosinophils. Meaning: GWAS Catalog: Eosinophile count (Sakaue S 2021; p=2e-159); T ↓ allele
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Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IL4R	rs1801275 16:27363079 · A/G/T	AG	chip	—	weak	IL4R Q576R: inconsistent associations with atopy. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FLG	rs61816761 1:152313385 · G/A/C/T	—	SNP not found in the data (Imputed with DR2=0.48 < 0.8)	A (—; 1%)	strong	FLG R501X: null mutation of filaggrin (skin barrier), risk factor of eczema, food allergy and asthma with eczema.

Histamine and drug response

Histamine is degraded by DAO (AOC1) in the intestine and by HNMT inside the cells. ADRB2 (Arg16Gly) affects the response to bronchodilators beta-2. Hereditary alpha-tryptasemia (extra TPSAB1 copies) is not read by chip.

Ageing mechanisms: Chronic Inflammation; Intestinal Barrier dysfunction

Favourable

None.

Unfavourable

• HNMT rs11558538 CT

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
HNMT	rs11558538 2:138002079 · C/T	CT	chip	T (1; 10%)	weak	HNMT Thr105Ile: T allele lowers the activity of HNMT (histamine intracellular degradation).

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
AOC1	rs10156191 7:150856517 · C/T	CT	chip	T (1; 26%)	weak	AOC1 (DAO) Thr16Met: T allele lowers intestinal DAO activity.
AOC1	rs1049793 7:150860577 · C/G/T	CG	chip	G (1; 27%)	weak	AOC1 (DAO) His645Asp: G allele lowers DAO activity.

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADRB2	rs1042713 5:148826877 · G/A/C	AG	chip	—	moderate	ADRB2 Arg16Gly Arg/Arg was associated with the worst response to regular use of salbutamol in some trials. Information.
ADRB2	rs1042714 5:148826910 · G/A/C/T	CG	imputation DR2=1.00	—	weak	ADRB2 Gln27Glu: small and inconsistent effect. Information.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Chronic Inflammation	TSLP CC; Chronic sinusitis (the 96th percentile); HNMT CT	Asthma (22 679 variants) (the 4th percentile); SMAD3 AA; C11orf30/EMSY GG	Asthma (212 loci) (the 31st percentile); Early childhood asthma (105 loci) (the 66th percentile); Asthma of onset in adults (the 45th percentile); ORMDL3/GSDMB CT; IL1RL1 CT; HLA-DQ TC; IL13 GG; IL2RA CC
Intercellular Communication	TSLP CC	Asthma (22 679 variants) (the 4th percentile); SMAD3 AA; Atopic disease in general (124 loci) (the 14th percentile); Allergic disease: asthma, rhinitis or eczema (136 loci) (the 22nd percentile); Atopic eczema (71 loci) (the 10th percentile); FCER1A TT	Asthma (212 loci) (the 31st percentile); Early childhood asthma (105 loci) (the 66th percentile); Asthma of onset in adults (the 45th percentile); ORMDL3/GSDMB CT; IL1RL1 CT; HLA-DQ TC; IL13 GG; IL2RA CC
Dysbiosis	Chronic sinusitis (the 96th percentile)	C11orf30/EMSY GG	Allergic rhinitis (349 variants) (the 52nd percentile); Nasal polyps (1783 variants) (the 39th percentile); ALOX15 GG
Intestinal Barrier dysfunction	HNMT CT	—	—
Remodelling of the extracellular matrix	TSLP CC	Asthma (22 679 variants) (the 4th percentile); SMAD3 AA	Asthma (212 loci) (the 31st percentile); Early childhood asthma (105 loci) (the 66th percentile); Asthma of onset in adults (the 45th percentile); ORMDL3/GSDMB CT; IL1RL1 CT; HLA-DQ TC; IL13 GG; IL2RA CC
Immunosenescence	—	Atopic disease in general (124 loci) (the 14th percentile); Allergic disease: asthma, rhinitis or eczema (136 loci) (the 22nd percentile); Atopic eczema (71 loci) (the 10th percentile); FCER1A TT	Eosinophils (607 loci) (the 58th percentile); Total IgE (7 loci) (the 46th percentile); Food allergy (peanut and others) (the 33rd percentile); IL5 TT

Practical part

Ideas for allergies and airways, crossed with the genetic plan v2 and current medication (isotretinoin, LDN, rapamycin, telmisartan, retatrutide) and with stack (vitamin D3, fish oil, zinc + copper). The current stack remains as it is; everything is suggestion (not applied). Without asthma or known allergies, none of this calls for treatment: they are signs to recognize early and low-cost care.

1. Synusite: early recognition

Genetics: Chronic sinusitis with p96 polyps; p39 polyps

Nasal congestion or thick discharge for more than 12 weeks, or loss of smell, request evaluation by otorhinolaryngology. Nasal washing with isotonic saline is safe and cheap in the days of crisis.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

2. Isotretinoin: dry nose

Genetics: No direct genetic relationship

Isotretinoin dry to nasal mucosa (crosses, bleeding, false sensation of rhinitis). Gel or spray of saline and humidifier in the fourth in winter solve most of them.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

3. Interior air and pollution

Genetics: Probable null GSTM1 (Phase I and II Detoxification report); GSTP1 Ile/Val; TSLP CC

The enzymes of the II phase that neutralize the oxidants of pollution are weak. Filter HEPA in the room, train away from traffic on days of ozone or high particles and cook with exhaust.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

4. Histamine: only if there are symptoms

Genetics: AOC1 2 heterozygous variants; HNMT Thr105Ile heterozygous

Keeping what the plan already says: without restriction of histamine except symptoms (redness, headache, diarrhea or nose clogged after cured cheeses, sausages or fermented). The copper of the stack (with zinc) is a cofactor of DAO.

V2 genetic plan and medication: Similar to the plan and the §12 profile. • *suggestion (not applied)*

5. If you ever have asthma

Genetics: ADRB2 Arg16Gly A/G; asthma p4-p66

Current first-line treatment (corticoid inhaled with formoterol) does not depend on ADRB2; regular salbutamol alone is not recommended in anyone.

V2 genetic plan and medication: No conflict. • *suggestion (not applied)*

6. Vitamin D and fish oil: already covered

Genetics: Allergic disease p22; eczema p10

Vitamin D and Omega-3 have modest evidence in the prevention of asthma attacks and allergic inflammation; they are already in the stack, without the need to change the dose due to these genes.

V2 genetic plan and medication: The same as the plan. • *suggestion (not applied)*

Avoiding or with caution

- Nasal decongestants in spray more than 5–7 days in a row (drug rhino). *suggestion (not applied)*
- Aspirin and other NSAID if nasal polyps with asthma appear one day (possible respiratory disease exacerbated by aspirin). *suggestion (not applied)*
- Dietary restrictions of histamine without symptoms. *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. HLA is read by SNP tag; alpha-tryptasemia (TPSAB1 copies) is not read.
- Asthma scores and atopic disease come from large GWAS; food allergy, medication and sinusitis have few loci. Histamine (AOC1, HNMT) and ADRB2 variants have weak to moderate evidence.
- Exposure (allergens, pollution, infections, tobacco) weighs much more on adult asthma than genes.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001849 (PGS Catalog; Penalized regression (bigstatsr); Asthma; PGP000263, PMID 34995502 · PGS004252 (PGS Catalog; Genome-wide signifier SNPs; Asthma; PGP000544, PMID 37080133 · PGS002248 (PGS Catalog; PRSice (C+T); Childhood asthma; PGP000275, PMID 35055391 · PGS001259 (PGS Catalog; snpnet; Hayfever/allergic rhinitis; PGP000244, PMID 35324888 · PGS001848 (PGS Catalog; Penalized regression (bigstatsr); Nasal polyps; PGP000263, PMID 34995502 · PGS003458 (PGS Catalog; Genome-wide signifier SNPs; General atopic disease; PGP000451, PMID 36804406 · PGS000810 (PGS Catalog; Variantly associated with allergic disease (asthma, hay fever or eczema); Allergic disease (asthma, hay fever or eczema); PGP000195, PMID 29083406 · PGS003486 (PGS Catalog; Genome-wide signifier SNPs; Atopic eczema; PGP000451, PMID 36804406 · PGS000165 (PGS Catalog; Genome-wide signifiers; Eosinophil count; PGP000078, PMID 32888494 · PGS000315 (PGS Catalog; GWAS-meaning (lead and secondary) variants; Immunoglobulin E (IgE); PGP000092, PMID 32527150).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

Generated in 02-10-2026 (local time) by the scripts of this report.